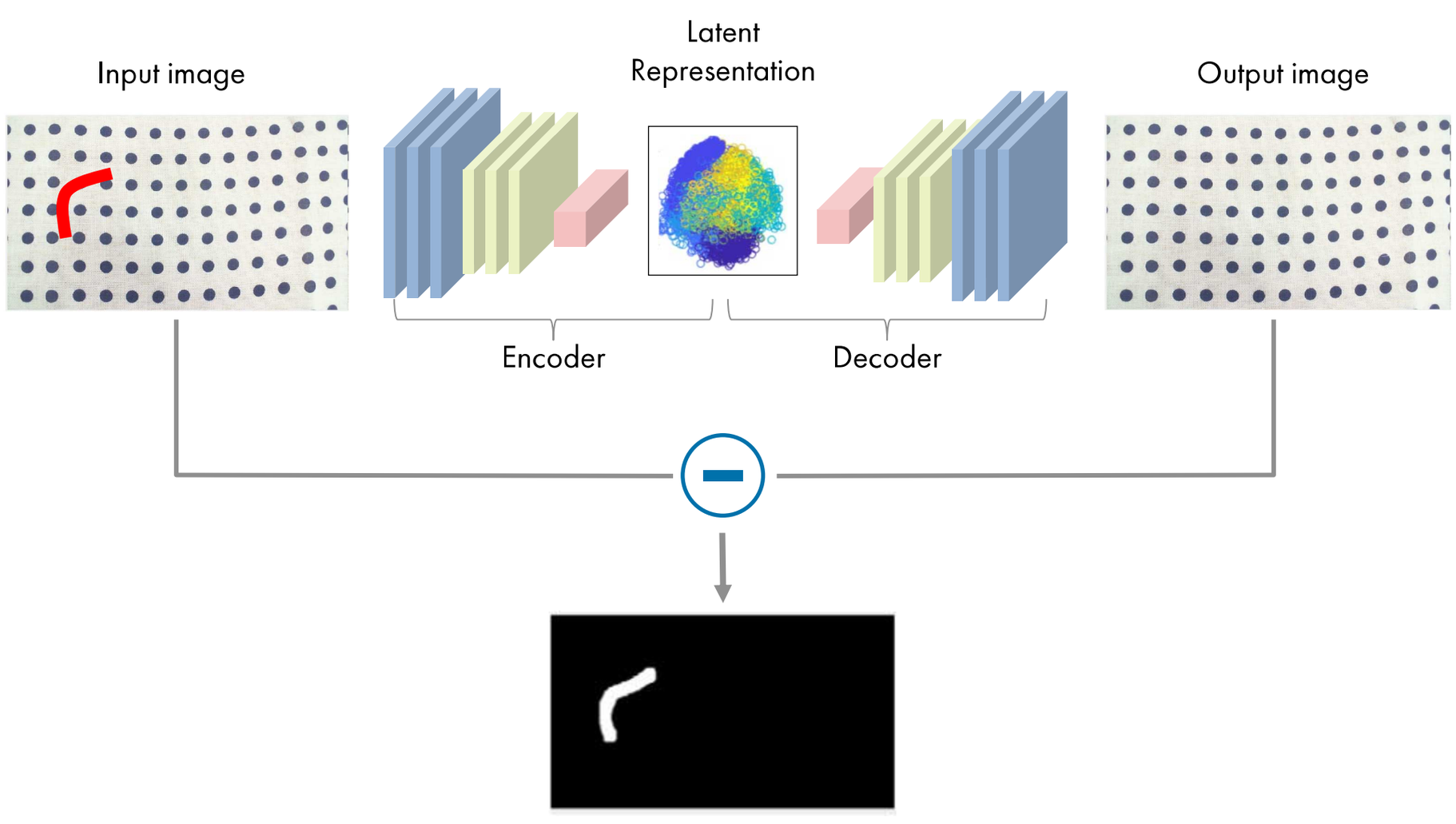
Lecture 1

Variational Autoencoders (VAE): from Autoencoders to Generative Models

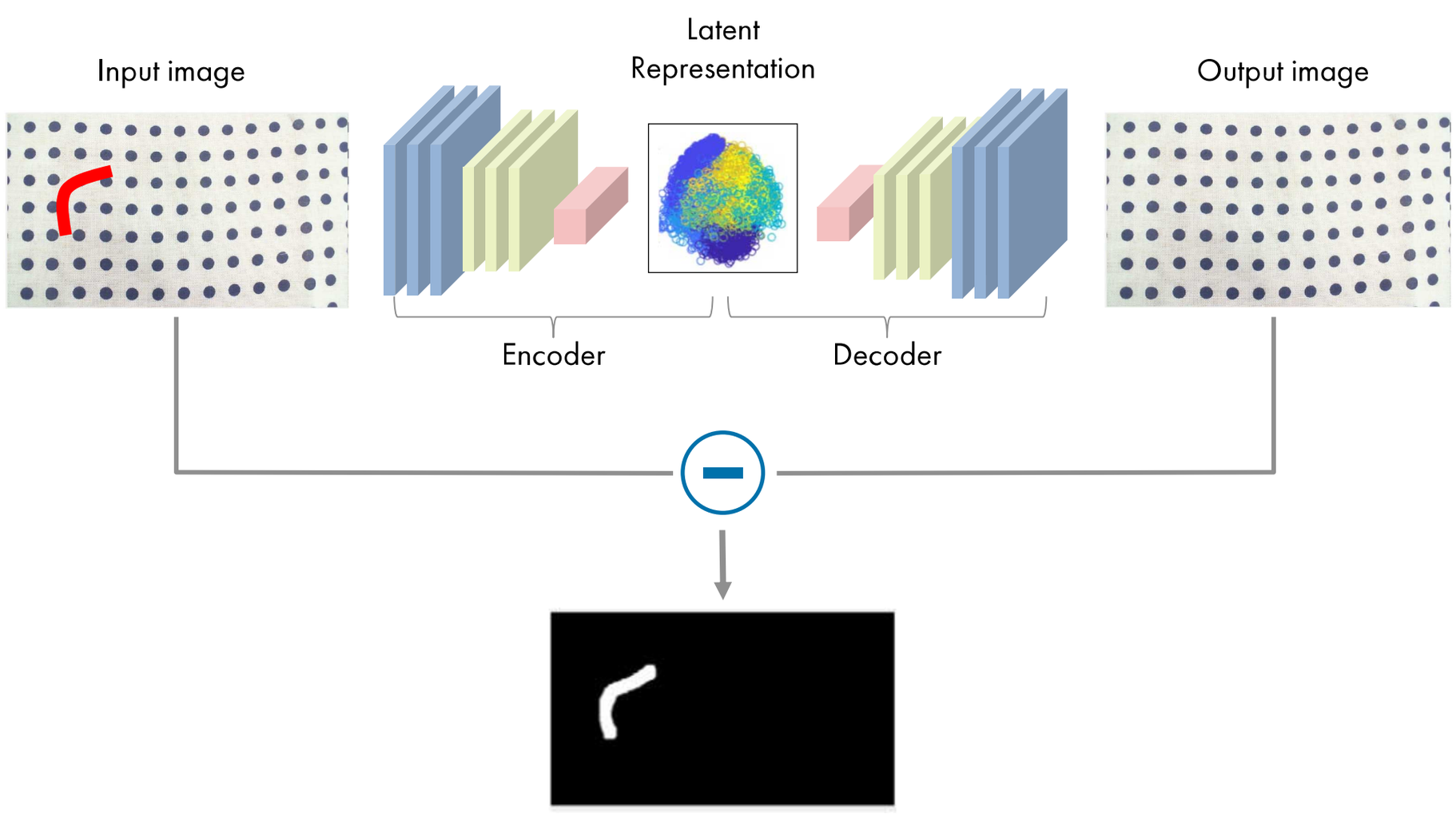
What is an Autoencoder?

An Autoencoder (AE) is a type of artificial neural network used primarily for learning efficient data representations (encoding) and reconstructing the input from this representation (decoding).



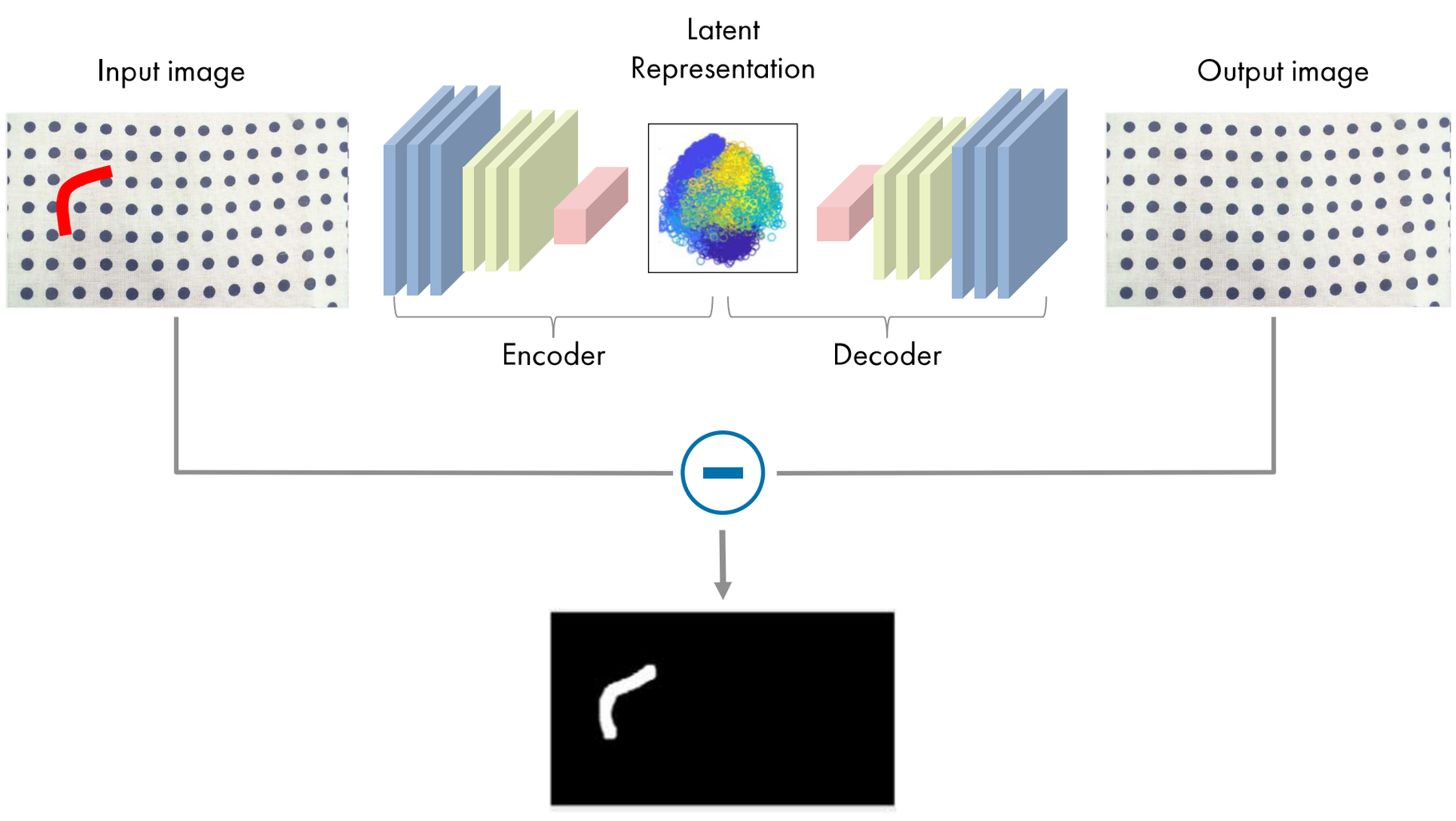
What is an Autoencoder?

- **Core Idea:**
  - Compress input data into a lower-dimensional latent space.
  - Reconstruct the original data from this compressed representation.
  - Evaluate performance using a **reconstruction loss**, typically mean squared error (MSE).



What is an Autoencoder?

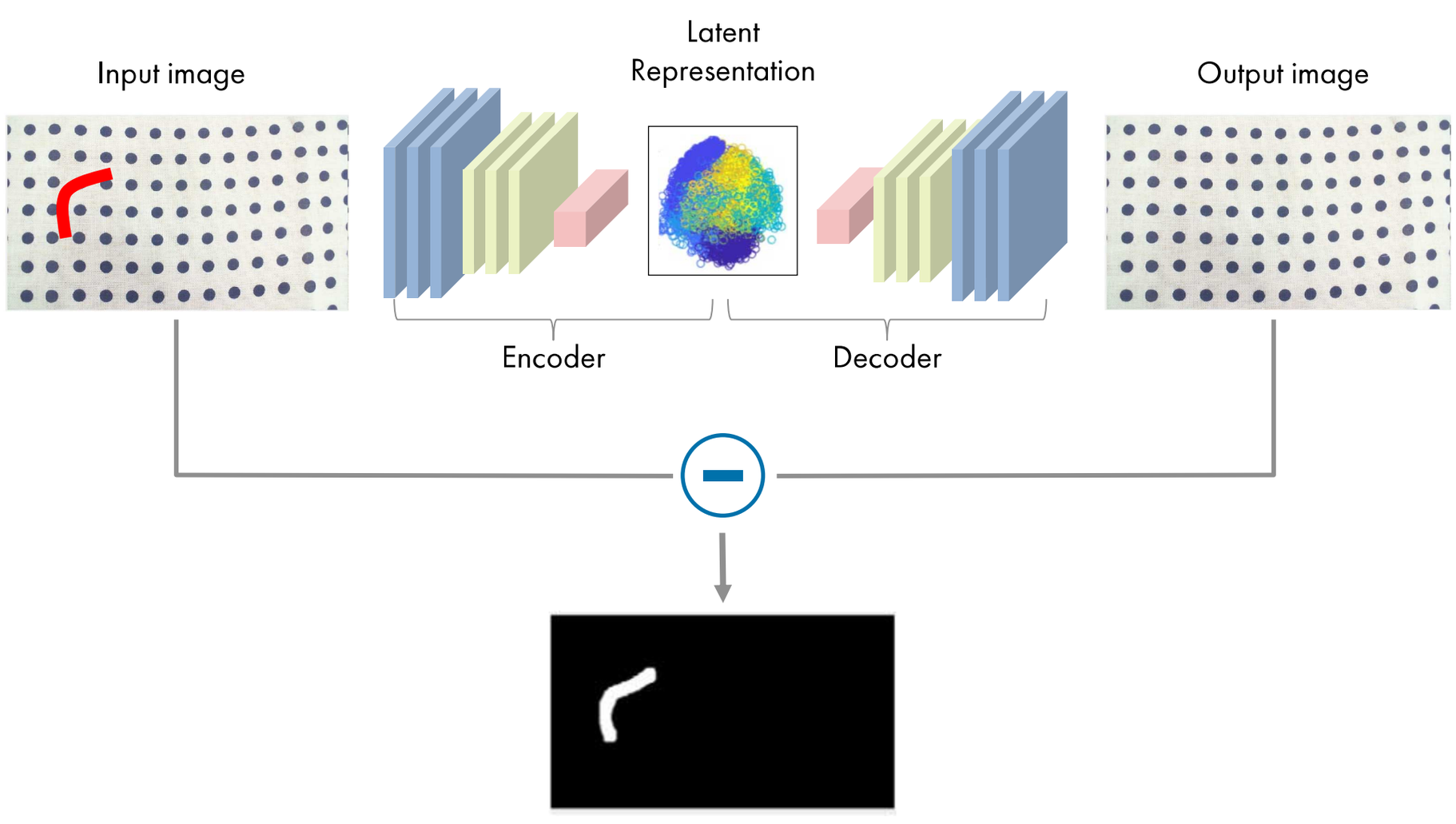
- **Historical Context:**
  - Introduced in the 1980s (Rumelhart, Hinton & Williams, 1986).
  - Gained popularity with deep learning advancements and powerful hardware (around 2010 onwards).



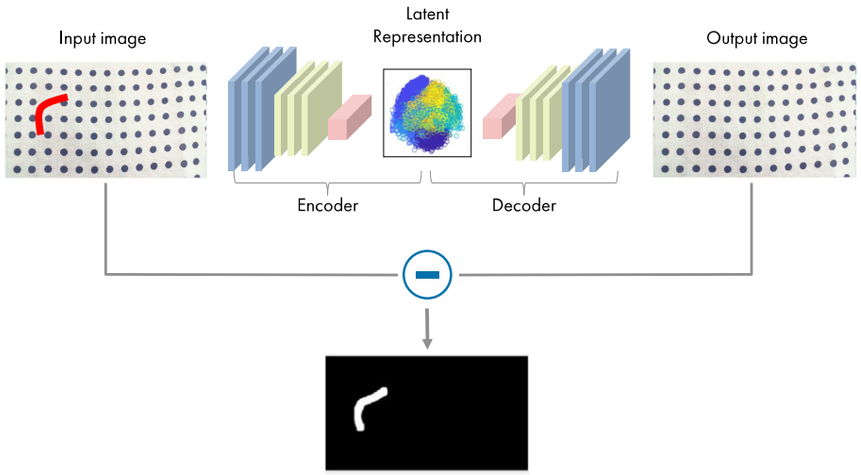
What is an Autoencoder?

General Reference (for students' further reading):

- **Goodfellow, I., Bengio, Y., & Courville, A.** (2016). *Deep Learning*. MIT Press.
  Chapter 14: Autoencoders (pp. 493–517).



Architecture of an Autoencoder



1. Encoder:

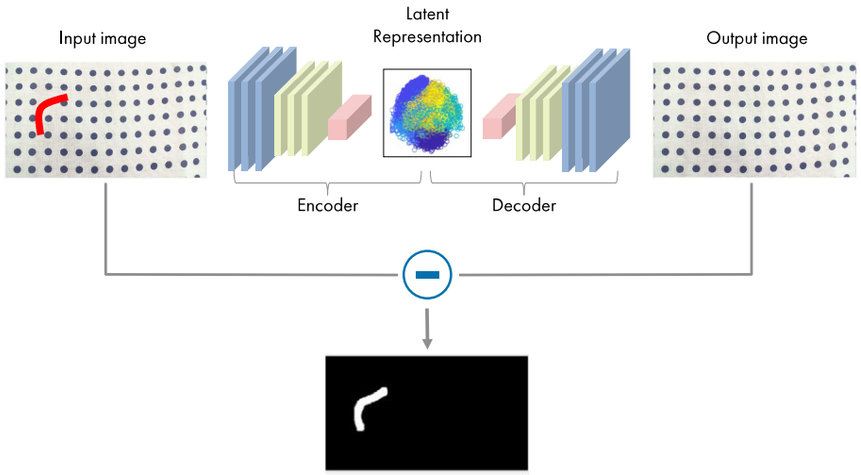
- A neural network that compresses the input data into a smaller latent representation.
- Mathematically, it performs the mapping:

$$z = f_\theta(x)$$

where:

- x is the input data.
- z is the latent representation.
- $f_\theta(\cdot)$ is the encoding function, parametrized by weights θ.

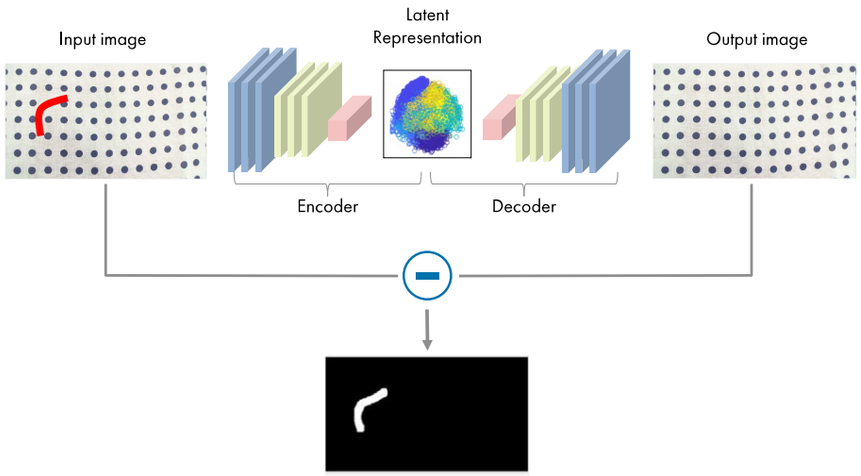
Architecture of an Autoencoder



2. Latent Space (Bottleneck Layer):

- A compressed representation capturing the essential features of the data.
- Lower dimensionality than the original input.

Architecture of an Autoencoder



3. Decoder:

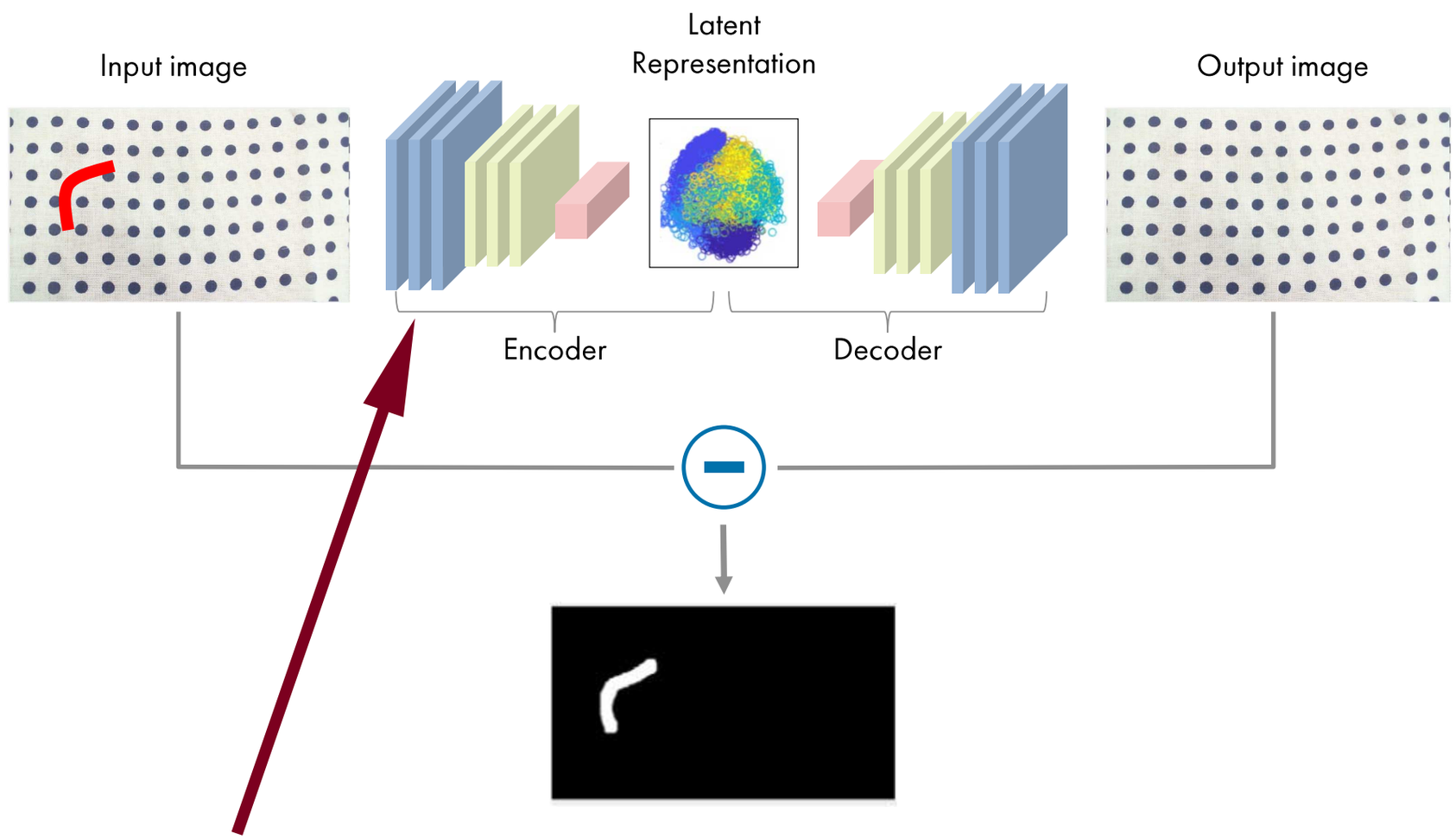
- A neural network that reconstructs data from the latent representation back into the original input space.
- Mathematically, it performs the mapping:

$$\hat{x} = g_\phi(z)$$

where:

- $\hat{x}$ is the reconstructed input.
- g_ϕ is the decoder network with parameters ϕ.

Architecture of an Autoencoder



Additional Notes (if needed for clarity):

- The encoder and decoder can be simple fully connected layers or more complex architectures (CNNs, RNNs), depending on the data type (images, sequences, etc.).
- The latent space dimension choice is crucial: too small loses information; too large might not learn useful representations.

Training an Autoencoder

Objective of Training:

- Minimize the difference between input x and reconstructed output $\hat{x}$.

Loss Function (Reconstruction Loss):

- Typically Mean Squared Error (MSE):

$$\mathcal{L}(x, \hat{x}) = \|x - \hat{x}\|^2 = \|x - g_\phi(f_\theta(x))\|^2$$

Training Process:

- The autoencoder learns by adjusting the weights of encoder (θ) and decoder (ϕ) using gradient descent:

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Training Process:

- The autoencoder learns by adjusting the weights of encoder (θ) and decoder (ϕ) using gradient descent:

- Training is unsupervised; no explicit labels are required.
- Hyperparameters (latent dimension size, learning rate, etc.) significantly affect model performance.

- Non-generative: cannot sample new data easily.

Limitations of Autoencoders

- **No Probabilistic Interpretation:**
  - The latent space is **deterministic**, meaning each input is mapped to a single latent vector, not a distribution.
- **Unstructured Latent Space:**
  - Latent space lacks continuity or smooth structure, making interpolation or sampling difficult or unreliable.
- **Limited Generative Capability:**
  - Autoencoders cannot easily generate new, realistic samples by randomly sampling the latent space.
- **Risk of Overfitting:**
  - Potentially memorizing the data rather than learning generalizable latent representations, limiting generative or exploratory analysis.

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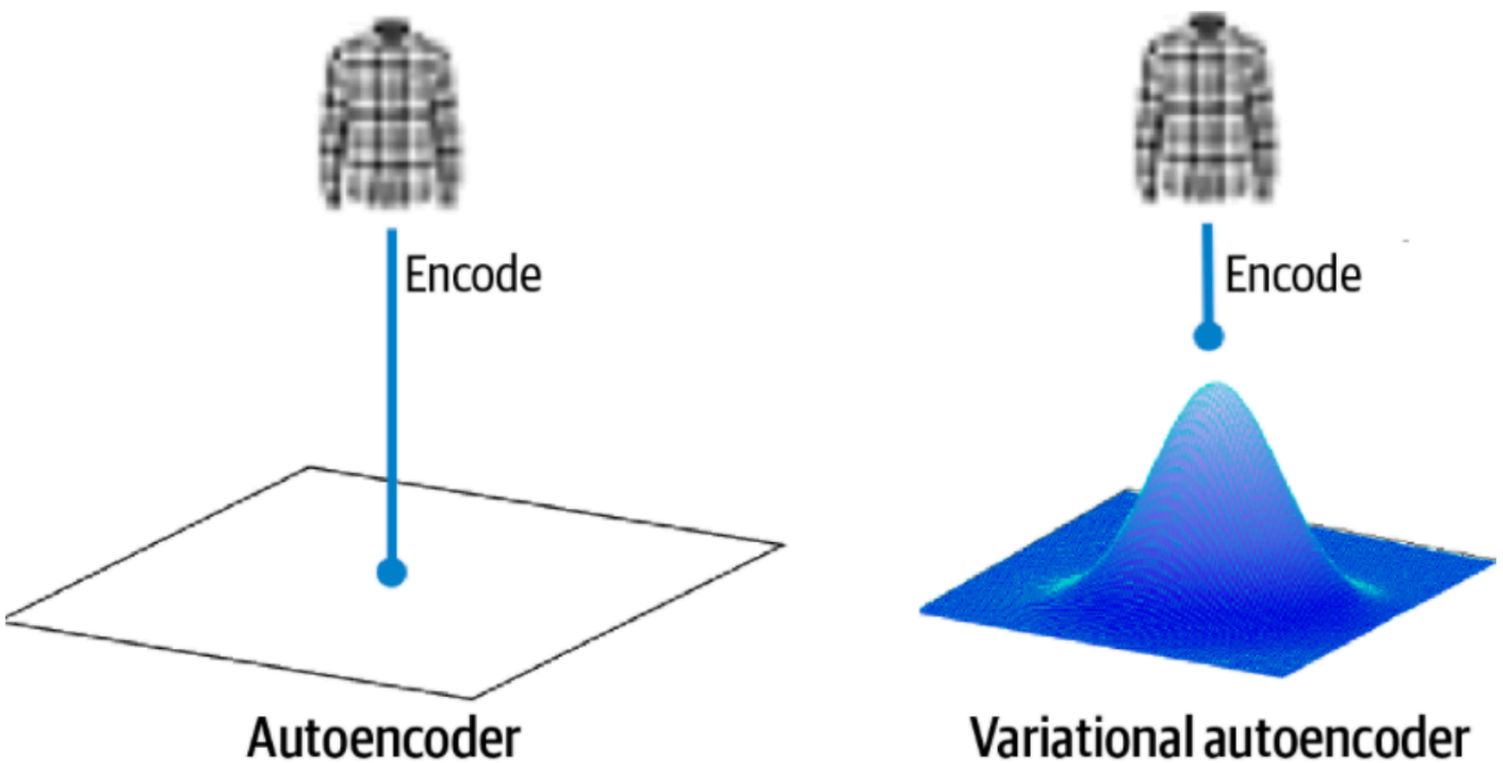

We turn to coding.....

AutoEncoder_3.ipynb

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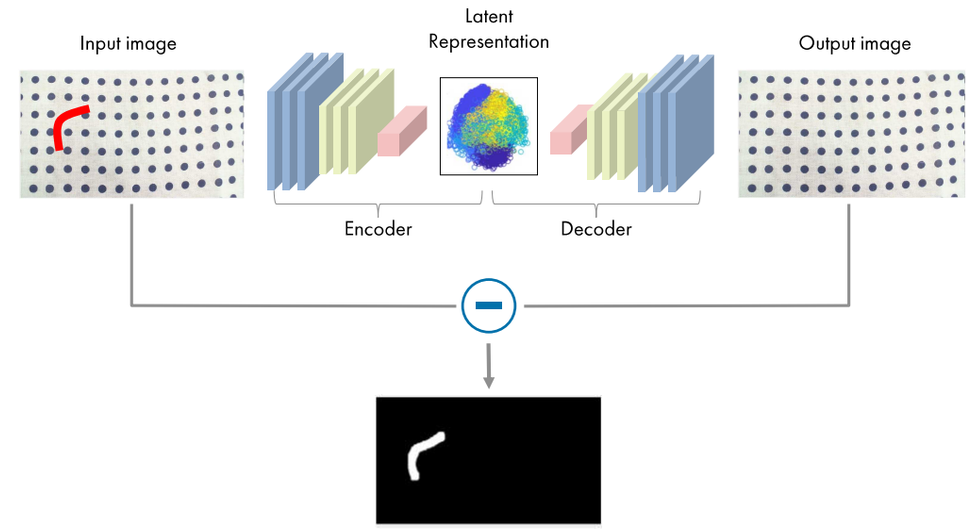
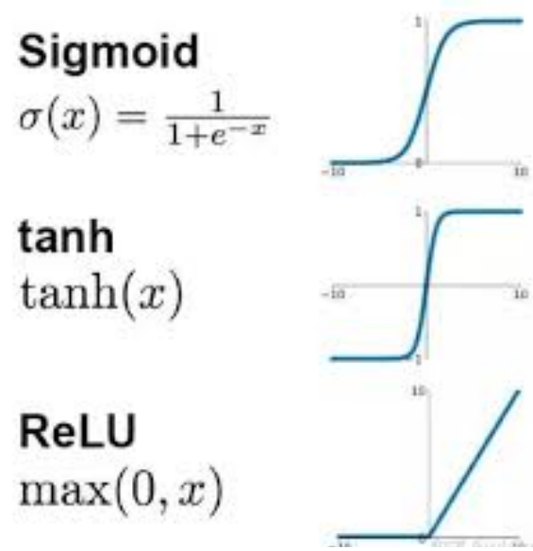


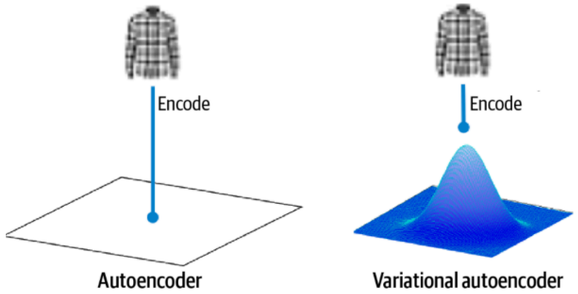


Linear

CLASS torch.nn.Linear(*in_features*, *out_features*, *bias=True*, *device=None*, *dtype=None*) [SOURCE]

Applies an affine linear transformation to the incoming data: $y = xA^T + b$.

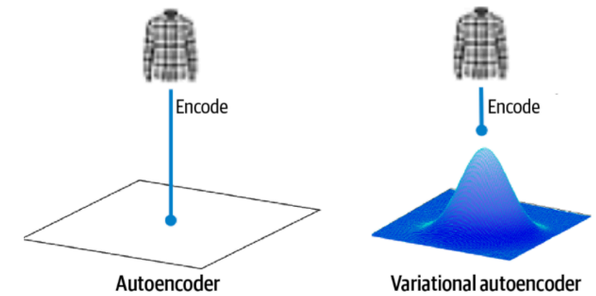
Motivation for Variational Autoencoders



Traditional Autoencoders face several critical limitations, motivating the development of **Variational Autoencoders (VAEs)**:

- **Structured Latent Spaces:**
  - **Autoencoders** often produce **irregular** and discontinuous latent spaces.
  - **Variational Autoencoders** encourage structured and continuous latent representations.
- **Generative Capabilities:**
  - VAEs explicitly define a **probability distribution** over latent space, enabling the generation of new data by sampling.
- **Probabilistic Interpretation:**
  - Allow quantifying **uncertainty** and variability, leading to more robust and interpretable latent features.
- **Smooth Interpolation:**
  - Smooth transitions between points in latent space facilitate meaningful data **interpolation and exploration**.

Introduction to Variational Autoencoders (VAE)



- **Definition:**

  > A **Variational Autoencoder (VAE)** is a generative model that extends the Autoencoder framework by introducing a probabilistic latent space. It encodes data into a probability distribution, allowing for the generation of new data samples.

- **Key Idea:**
  - **Encoder** maps inputs not to fixed points, but to a probability distribution (usually Gaussian) in latent space.
  - **Decoder** generates new, plausible data points by sampling from this latent distribution.